Economics in Action 1
Answers to Exercises — Exam-type questions

Chapter 5 Production in the short run and long run

Questions

pp. 30–35

1.

A supermarket plans to extend its opening hours to 24 hours. Which of the following are variable factors involved in the extra opening hours?

(1) The premises
(2) Full-time accounting clerk
(3) Night shift security guard
(4) Electricity
(5) Delivery lorry
(6) Part-time shop assistants

A. (1), (2) and (5) only
B. (3), (4) and (6) only
C. (2), (3), (4) and (6) only
D. (1), (3), (4), (5) and (6) only

2.

John is a taxi owner who drives his own taxi. Which of the following are his fixed costs of providing taxi service?

(1) The taxi
(2) Annual licence fee of the taxi
(3) His human effort as a driver
(4) Fuel cost
(5) Insurance cost

A. (1) and (3) only
B. (2) and (5) only
C. (1), (2) and (5) only
D. (2), (4) and (5) only

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3.

The following table shows the input-output relationship of a firm which employs capital and labour only.

Capital (units)	Labour (units)	Marginal product (units)
2	1	6
2	2	7
2	3	5
2	4	2
2	5	-1

The average product of labour is the highest when the quantity of labour is _____ units, while the total product is the highest when the quantity of labour is _____ units.

A. 2 …… 3
B. 2 …… 4
C. 3 …… 3
D. 3 …… 4

4.

The input-output relationship of a firm is shown in the table below. It only employs two factors, namely capital and labour.

Capital (units)	Labour (units)	Total output (units)
5	2	90
5	3	160
5	4	270
5	5	360

According to the table above, diminishing marginal returns set in when the _____ unit of labour is employed.

A. 2nd
B. 3rd
C. 4th
D. 5th

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5.

The following table shows the average output of a firm. The only inputs required are labour and machines.

Machines (units)	Labour (units)	Average product (units)
2	2	13
2	3	14
2	4	15
2	5	16
2	6	X

If the law of diminishing marginal returns applies to the firm, the value of X

A. must be smaller than 16.
B. must be larger than 16.
C. can be any value greater than 16.
D. can be any value smaller than 16.

6.

The following table shows the production plan of a firm which only employs two factors, namely land and labour.

Land (units)	Labour (units)	Total product (units)
2	3	30
3	3	45
4	3	65
5	3	X
6	3	Y

Suppose the above data illustrate the law of diminishing marginal returns. Which of the following figures are consistent with the law?

(1) X = 65, Y = 45
(2) X = 90, Y > 110
(3) X > 93, Y = 121

A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only

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7.

When a firm only increases the amount of capital to increase its output,

A. labour productivity will decrease.
B. the optimal scale of production will decrease.
C. the average cost of production will eventually fall as a result of economies of scale.
D. the marginal product of capital will eventually decrease.

8.

Which of the following statements about economies of scale is correct?

A. Only large firms enjoy economies of scale.
B. This is caused by the law of diminishing marginal returns in the long run.
C. It begins after a firm's output exceeds the optimal scale of production.
D. It can be the result of spreading the production cost over a larger output.

9.

The following table shows the production data of a firm.

Output (units)	0	1	2	3	4
Total cost ($)	8	15	20	24	36

According to the above table, we can conclude that

A. the optimal scale of production is 3 units of output.
B. the firm enjoys economies of scale when it increases output from 2 units to 3 units.
C. the total variable cost is $12 when output is 2 units.
D. the average fixed cost does not vary with output.

10.

The following table shows the production plan of a firm which only employs two factors, labour and capital.

Workers (units)	Average working hours per worker	Capital (units)	Average cost ($)
20	1	20	15
20	2	30	25
20	3	40	35
20	4	50	45
20	5	60	55

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The above table shows a _____ situation in which _____.

A. short run …… the law of diminishing marginal returns does not apply
B. short run …… the law of diminishing marginal returns applies
C. long run …… economies of scale exist
D. long run …… diseconomies of scale exist

11.

The table below shows the total output of ABC ice cream shop per day. It employs only two inputs in production.

Premises (units)	2	2	2	2
Number of workers	10	10	10	10
Average working hours per worker	7	8	9	10
Total output (units)	3,400	3,700	3,900	4,000

As the market demand for ice cream increases, ABC ice cream shop raises its output by requiring workers to work longer hours, from 7 hours to 10 hours per day. Which of the following is correct?

A. The fixed cost is zero because the production cost of premises is zero.
B. The amount of labour employed by the firm remains unchanged.
C. The firm enjoys external economies of scale.
D. The change in output can be explained by the law of diminishing marginal product.

12.

As the government implements policies to attract more IT companies to set up offices in the city, a small local IT company has to increase its salaries by 10% in order to hire graduates with an IT degree. The local company

A. enjoys internal economies of scale.
B. suffers internal diseconomies of scale.
C. enjoys external economies of scale.
D. suffers external diseconomies of scale.

13.

When a producer reduces all its inputs to cut its output by 50%, the total cost drops by 60%. Which of the following is a possible reason?

A. The law of diminishing marginal returns applies to the firm.
B. The organisation of the firm was too complicated before the reduction in output.
C. The interest rate of borrowing from banks increases after the reduction in output.
D. The firm can utilise its machines more fully after reducing its production scale.

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14.

A fast-food shop opens inside a theme park.

(a) Give TWO examples of variable costs incurred in the operation of the fast-food shop. Explain your answer. (3 marks)

(b) Give TWO examples of fixed costs incurred in the operation of the theme park. Explain your answer. (3 marks)

15.

The input-output relationship of a firm is shown below. The only inputs required are labour and tools.

Labour (units)	Tools (units)	Total product (units)
20	2	50
21	3	55
22	4	62
23	5	70
24	6	76

State the law of diminishing marginal returns and explain whether the above data illustrate the law. (5 marks)

16.

The following table shows the average product of a firm.

Labour (units)	Capital (units)	Average product (units)
10	2	200
10	3	250
10	4	350
10	5	500
10	6	595

Explain the economic principle illustrated by the data above. (5 marks)

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17.

Ken runs a bakery. Despite an influx of tourists during the summer, he is unable to hire more workers and can only purchase more tools to increase the daily production of bread.

(a) Does Ken's bakery operate in the short run or long run? Explain your answer. (2 marks)

(b) After purchasing more tools, Ken found that the marginal product of tools has dropped. Which economic law can explain the above phenomenon? State the law and explain how it can be applied in this case. (6 marks)

18.

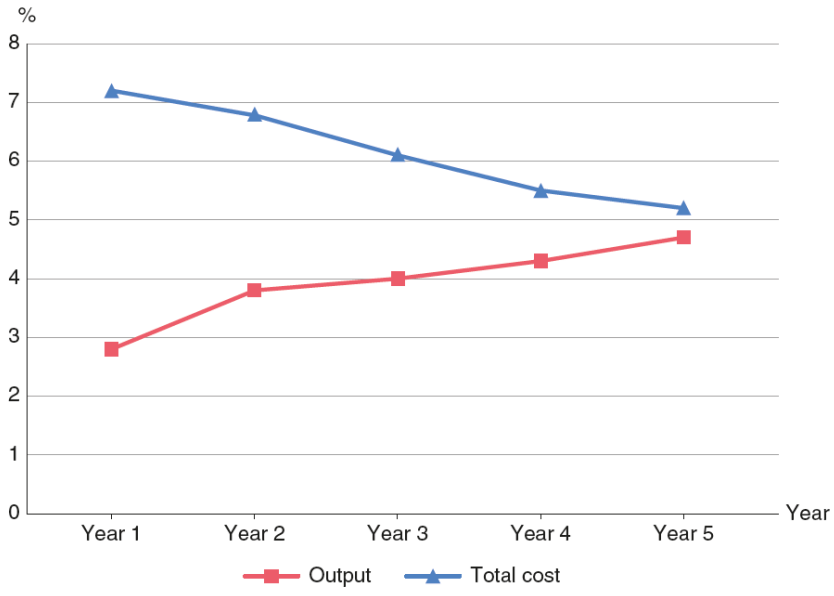
After a bubble tea shop opened a new branch, the average cost of production decreased. Explain THREE possible reasons for such a change in average cost. (6 marks)

19.

There are many cafés clustered in Sham Shui Po. In terms of average cost, explain TWO reasons for the phenomenon. (4 marks)

20.

The CEO of an electric vehicle manufacturer implemented a plan to increase its production by expanding all inputs, including workers, factories and machines. The results of the plan are shown in the diagram below.



(a) Did the electric vehicle manufacturer operate in the short run or long run? Explain. (2 marks)

(b) Based on the above diagram, explain how the average cost has changed from Year 1 to Year 5. (2 marks)

(c) Explain the change in average cost with TWO possible economic reasons. (4 marks)

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Answers

1. B

2. B

3. B

Capital (units)	Labour (units)	MP (units)	TP (units)	AP (units)
2	1	6	6	6
2	2	7	13	6.5
2	3	5	18	6
2	4	2	20	5
2	5	-1	19	3.8

4. D

Capital (units)	Labour (units)	Total output (units)	MP
5	2	90	-
5	3	160	70
5	4	270	110
5	5	360	90

5. D

Machines (units)	Labour (units)	AP (units)	TP (units)	MP (units)
2	2	13	26	-
2	3	14	42	16
2	4	15	60	18
2	5	16	80	20
2	6	X	6X	6X – 80

$6X - 80 < 20$

$X < 100 \div 6$

$X < 16.7$

6. C

	MP of the 5th unit (= X − 65)	MP of the 6th unit (= Y − X)
(1)	0	-20
(2)	25	> 20
(3)	> 28	< 28

7. D

8. D

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9. C

When output is zero, TC is $8, implying that TFC is $8. The production is in the short run.

10. D

11. D

12. D

13. B

After reducing production scale, average cost of production decreases.

14.

(a) Variable costs: electricity fee, cost of food, wages of part-time staff

(Any TWO of the above or other reasonable answers. 1 mark × 2 = 2 marks)

These expenses will increase with the output of the fast-food shop. (1 mark)

(b) Fixed costs: rent for land, insurance premium, business registration fee

(Any TWO of the above or other reasonable answers. 1 mark × 2 = 2 marks)

These expenses will not increase with the output of the theme park. (1 mark)

15.

The law of diminishing marginal returns states that in the short run, when more units of a variable factor are continuously added to a given quantity of fixed factors, the marginal product of the variable factor will eventually decrease, holding technology constant. (3 marks)

The data do not illustrate the law because all factors are variable. (2 marks)

16.

The data illustrate the law of diminishing marginal returns. (1 mark)

When more units of capital (variable factor) are continuously added to a given quantity of labour (fixed factor), the marginal product of capital eventually decreases from 1,100 units to 1,070 units, ceteris paribus (including technology). (4 marks)

Labour (units)	Capital (units)	Average product (units)	Total product (units)	Marginal product (units)
10	2	200	400	-
10	3	250	750	350
10	4	350	1,400	650
10	5	500	2,500	1,100
10	6	595	3,570	1,070

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17.

(a) It operates in the short run. (1 mark)

Workers are a fixed factor as their quantity does not change with output. (1 mark)

(b) The law of diminishing marginal returns. (1 mark)

The law states that in the short run, when more units of a variable factor are continuously added to a given quantity of fixed factors, the marginal product of the variable factor will eventually decrease, holding technology constant. (3 marks)

In this case, the tools are the variable factor and workers are the fixed factor. The tools may become idle and not fully used so the marginal product may fall. (2 marks)

18.

Reasons:

- The shop can have more extensive division of labour among its employees. This increases managerial efficiency and productivity.
- The shop may borrow money from banks at a lower interest rate because more collateral will be available for a larger firm.
- The shop can buy raw materials in bulk and obtain a larger discount.
- The shop can spread its marketing costs over a larger output.
- The shop can spread its research and development costs over a larger output.
- The shop can utilise machines and equipment more fully, increasing productivity.

(Any THREE of the above or other reasonable answers. 2 marks × 3 = 6 marks)

19.

Reasons:

- Lowering the costs of marketing and promotion as the region attracts café customers.
- Lowering transportation cost for suppliers (e.g., food ingredients and coffee beans).
- Lowering the costs of sourcing as more back-up services (e.g., maintenance and cleaning) are available.

(Any TWO of the above or other reasonable answers. 2 marks × 2 = 4 marks)

20.

(a) Long run because all factors of production are variable. (2 marks)

(b) Average cost increases because the percentage increase in total cost is greater than the percentage increase in output. (2 marks)

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(c) Reasons:

- The manufacturer's organisation becomes too complicated, lowering managerial efficiency.
- If the manufacturer already has large outstanding loans, it may need to pay a higher interest rate to obtain more loans.
- The manufacturer may need to purchase more expensive inputs and use more expensive substitutes as the supply of necessary inputs is limited.
- The manufacturer may need to incur higher marketing costs to promote its products as the market has become saturated.

(Any TWO of the above or other reasonable answers. 2 marks × 2 = 4 marks)

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